Chapter 7
Fluid Pressure, Fluid Flow in the Body, and Motion in Fluids

Abstract The basics of fluid pressure, volume, and flow and how they are interrelated are presented in this chapter to set the stage for the analysis of fluids in later chapters: that of blood in the circulatory system and air in the respiratory system. The concepts of fluid resistance and compliance are developed, along with those of drag and lift on objects flowing in fluids and diffusion in fluids. Motion of humans in fluids is also discussed.

In the following two chapters we will examine the flow of two fluids in the body: blood in the heart and circulatory system, and air in the lungs and respiratory system. Flow of fluids elsewhere in the body is also important, such as urine in the renal and urinary systems, and food and digestive juices in the gastrointestinal system [20, 23, 28]. Such directed flow of material in the body occurs predominantly by fluid flow in systems of vessels. Directed transport also occurs on microtubules in the body by motor proteins, such as kinesin and dynein, as described in Chap. 5. Undirected motion occurs by diffusion, which can be passive (including by osmosis) or active (assisted by energy or by pressure differences) [28]. Over "longer" distances such direct transport is preferred to diffusion because it provides a directed motion and a motion that is faster than diffusion. Diffusion is important in the body only over very short distances, up to $\sim 100\,\mu$m.

In this chapter we will discuss the concept of pressure as it relates to fluids in the body. For example, the pressure of the vitreous humor in the eyeball serves several functions, including maintaining the shape of the eyeball. This pressure is similar to the stress we examined in Chap. 4, such as that in our long bones when we walk. They both describe a force per unit area. The pressure in the fluid is hydrostatic, i.e., the force per unit area is the same stress in all directions. In solids the stress is often anisotropic. We will review the basic physics of pressure and fluid flow, including the relationship of pressure and fluid flow, and diffusion [11, 20, 23, 28]. We will also examine the flow of humans in fluids, i.e., swimming, along with the possibility of human flight.

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7.1 Characteristic Pressures in the Body

7.1.1 Definition and Units

The pressure of a fluid column is given by (2.48), $P = \rho gh$, where ρ is the fluid density, g is the gravitational constant, and h is the height of the column. For mercury ρ is 13.6 g/cm^3. For water $\rho = 1.00$ g/cm^3 at 4 °C. The density of whole blood is a bit higher, 1.06 g/cm^3 at 37 °C. The units of pressure are presented in Table 2.6.

So far we have been discussing *absolute pressure*, P_{abs}, which is the total force per unit area. In discussions concerning the body it is very common to cite the *gauge pressure*, P_{gauge}, which is the pressure relative to a standard, which is usually atmospheric pressure, and so $P_{\text{gauge}} = P_{\text{abs}} - 1$ atm. This is helpful because it is the difference in pressure that is the net force that acts on a unit area. In discussing blood pressure and the pressure of air in the lungs, it is assumed that the term pressure P refers to the gauge pressure relative to the local atmospheric pressure. During breathing in (which is called inspiration), the pressure in the lungs is lower than that outside the body and so the internal (gauge) pressure is <0. Table 7.1 gives typical pressures in the body.

7.1.2 Measuring Pressure

One way of directly measuring pressure is with a manometer (Fig. 7.1). The measured pressure is that corresponding to the height of the fluid column plus the reference pressure, so

$$P = P_{\text{ref}} + \rho gh. \tag{7.1}$$

The most common way to measure blood pressure is with a *sphygmomanometer* (sfig-muh-ma-nah'-mee-ter), which consists of a cuff, a squeeze bulb, and a meter that measures the pressure in the cuff (Fig. 7.2). The cuff is the balloon-like jacket placed about the upper arm above the elbow; this encircles the brachial artery. The cup of a stethoscope is placed on the lower arm, just below the elbow, to listen for the flow of blood. With no pressure in the cuff, there is normal blood flow and sounds are heard through the stethoscope. Gurgling sounds are heard after the cuff is pressurized with the squeeze bulb and then depressurized by releasing this pressure with a release valve in this bulb.

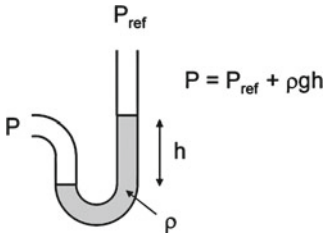
To understand when these sounds occur and their significance, we need to understand how the pressure in the main arteries varies with time. (This will be detailed in Chap. 8.) In every heart beat cycle (roughly 1/s), the blood pressure in the major arteries, such as the brachial artery, varies between the systolic pressure ($\sim$120 mmHg) and the diastolic pressure ($\sim$80 mmHg), as is depicted in Fig. 7.3. (The units of these cited gauge pressures are in mmHg—see (7.1) and Chap. 2.) When the pressure in the cuff exceeds the systolic pressure, there is no blood flow to the lower arm and

Table 7.1 Typical (gauge) pressures in the body (in mmHg)

Arterial blood pressure	
Maximum (systolic)	100–140
Minimum (diastolic)	60–90
Capillary blood pressure	
Arterial end	30
Venous end	10
Venous blood pressure	
Typical	3–7
Great veins	<1
Middle ear pressure	
Typical	<1
Eardrum rupture threshold	120
Eye pressure	
Humors	20 (12–23)
Glaucoma threshold range	~21–30
Cerebrospinal fluid pressure	
In brain—lying down	5–12
Gastrointestinal	10–12
Skeleton	
Long leg bones, standing	~7,600 (10 atm.)
Urinary bladder pressure	
Voiding pressure	15–30 (20–40 cmH_2O)
Momentary, up to	120 (150 cmH_2O)
Intrathoracic	
Between lung and chest wall	−10

Using data from [3]

Fig. 7.1 Manometer



consequently there are no sounds. When the pressure in the cuff is lowered with the release bulb to just below the systolic pressure, there is intermittent flow. During the part of the cycle when the arterial blood pressure is lower than the cuff pressure there is no flow; when it is greater, there is flow. This intermittent flow is turbulent and produces gurgling sounds. These sounds, the *Korotkoff* or K *sounds*, are transmitted by the stethoscope. As the cuff pressure is lowered further, the K sounds get louder and

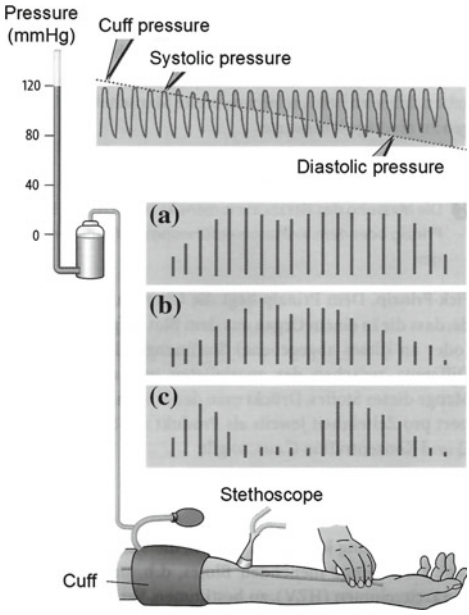


Fig. 7.2 Measuring blood pressure with a sphygmomanometer, listening to Korotkoff sounds (of varying levels during the turbulent flow shown in **a**–**c**). (Listening to sounds is called *auscultation*) (From [29])

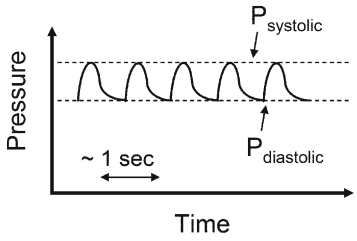


Fig. 7.3 Variation of blood pressure with time, for blood leaving the left heart for the systemic system, with the systolic and diastolic pressures shown

then lower, and are heard until the cuff pressure decreases to the diastolic pressure. Blood flow is not interrupted when the cuff pressure is less than the diastolic pressure and the K sounds cease because the blood flow is no longer turbulent. Therefore, the onset and end of the K sounds, respectively, denote the systolic and diastolic blood pressures. (This auscultatory method of Korotkoff was introduced by Russian army physician Korotkoff [24] who discovered a century ago that sound can be heard distally from a partially occluded limb [7].)

7.2 Basic Physics of Pressure and Flow of Fluids

In this section we overview the basics of fluids. Some of this will be a review for most. Some of the more advanced results are derived, while others are merely presented. These basics will be used in subsequent chapters.

Both gas and liquid fluids are important in the body. Gases will be treated by the ideal gas law

$$P = nRT, \tag{7.2}$$

where P is the pressure, n is the gas density, R is the gas constant (=8.31 J/mol-K), and T is the temperature (in K). The gas density $n = N/V$, where N is the total number of molecules in a volume V. The gas constant $R = N_A k_B$, where N_A is Avogadro's number, 6.02×10^{23}, and k_B is Boltzmann's constant, 1.381×10^{-23} J/K.

One guiding principle is *Pascal's Principle*: the pressure applied to a confined fluid increases the pressure throughout by the same amount. Also quite important is *Archimedes' Principle*: the buoyant force on a body immersed in a fluid is equal to the weight of the fluid displaced by that object. Another important relation is the *Law of Laplace*, which relates the difference of pressures inside and outside a thin-walled object—of a given shape—to the tension in the walls of the object. We will also need to understand the properties of flowing fluids to be able to analyze the physics of the circulatory system.

7.2.1 Law of Laplace

The pressure inside blood vessel walls, P, exceeds that outside, P_{ext}, by $\Delta P = P - P_{ext}$. *How large of a tension should the vessel walls be able to withstand to support this positive pressure difference in equilibrium?* The answer is provided by the *Law of Laplace* for hollow cylinders. It is derived here and then used in Chap. 8.

Consider a tube of radius R and length L. Figure 7.4a shows a section of this tube with angle $\theta \ll 1$. The outward force (upward in the diagram) on this area is the pressure difference, ΔP, times the area, $(R\theta)L$. The circumferential tension T is the force per unit length (along the tube length). (Note that this use of the word "tension" has a different meaning than in earlier chapters, where it meant a force, often used to pull things apart.) This film tension has units of force/length or energy/area. It is equal to a circumferential stress $\sigma = T/w$, where w is the cylinder thickness (with $w \ll R$). These forces can be those within the blood vessel walls (Chap. 8). The horizontal components of the film tension to the left and right cancel. The vertical components are inward and each equal to $T \sin(\theta/2) \simeq T(\theta/2)$ for small angles. With both of these tension components multiplied by L, in static equilibrium force balance gives

$$\Delta P(R\theta)L = 2\left(T\frac{\theta}{2}\right)L. \tag{7.3}$$

This means

$$\Delta P = \frac{T}{R} \quad \text{or} \quad T = R(\Delta P). \tag{7.4}$$

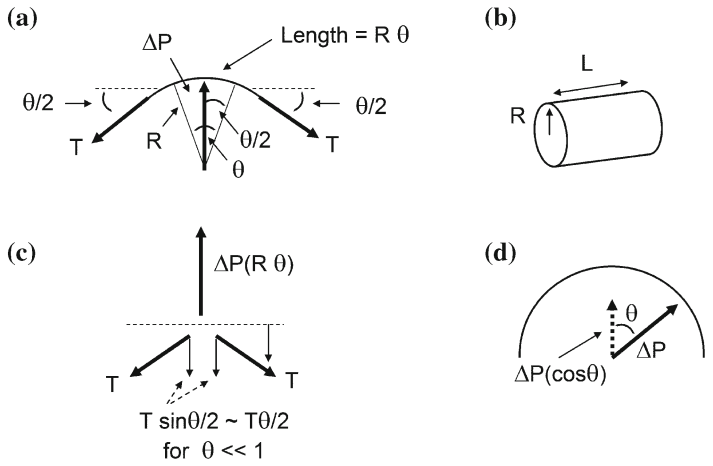


Fig. 7.4 Derivation of the Law of Laplace for the cylinder in (**a**), with the force diagram for a section of a cylinder in (**b**), leading to the force diagram in (**c**), and the resolution of pressures for analysis of a half cylinder in (**d**)

This is a differential method. Alternatively we could integrate the forces over a half cylinder, as shown in Fig. 7.4d. The total downward force is the area of the walls, $2wL$, times the stress, σ, or $2wL\sigma$. The total upward force is the cross-sectional area, $2RL$, times the pressure difference, ΔP, or $2RL(\Delta P)$. In equilibrium

$$2RL(\Delta P) = 2wL\sigma \tag{7.5}$$

$$\Delta P = \frac{w\sigma}{R} = \frac{T}{R}, \tag{7.6}$$

which is the same as (7.4). (Figure 7.4d shows that the total upward force is really the integral of the upward force component, $\Delta P \cos\theta$, times the area element, $RL\mathrm{d}\theta$, integrated from $-90°$ to $90°$ or

$$(\Delta P)RL \int_{-90°}^{90°} \cos\theta \mathrm{d}\theta = (\Delta P)RL(\sin(90°) - \sin(-90°)) = 2(\Delta P)RL, \tag{7.7}$$

which turns out to be the same as expected.)

The Law of Laplace is also important in spheres, such as soap bubbles and the alveoli in the lungs. For a sphere of radius R and wall thickness w, we can balance the forces in the half sphere. The total downward force is the area of the walls, $2\pi Rw$, times the stress, σ, or $2\pi Rw\sigma$. The total upward force is the cross-sectional area, πR^2, times the pressure difference, ΔP, or $\pi R^2(\Delta P)$. In equilibrium

$$\pi R^2(\Delta P) = 2\pi Rw\sigma \tag{7.8}$$

$$\Delta P = \frac{2w\sigma}{R} = \frac{2T}{R}. \tag{7.9}$$

Table 7.2 Surface tension (γ) for several liquids

Liquid	T (°C)	γ (10^{-4} N/m)
Water	0	7.56
	20	7.28
	60	6.62
	100	5.89
Whole blood	20	5.5–6.1
Blood plasma	20	5.0–5.6
Lung surfactant	20	0.1
Cerebrospinal fluid	20	6.0–6.3
Saliva	20	1.5–2.1
Benzene	20	2.89
Mercury	20	46.4

Using data from [9, 15, 26]

This is the Law of Laplace for a sphere. We will use it in Chap. 9. (It is derived in more detail in Problem 7.12.)

For a spheroid with different radii of curvature, R_1 and R_2, (7.4) and (7.9) generalize to

$$\Delta P = \frac{T}{R_1} + \frac{T}{R_2}. \tag{7.10}$$

For a cylinder, $R_1 = R$ and $R_2 = \infty$ and this reduces to (7.4). For a sphere, $R_1 = R$ and $R_2 = R$ and it reduces to (7.9).

Our force balance arguments have made a direct connection between this tension, or really surface tension, and its units of force/length. Surface tension also has the same units as energy/area. This is reasonable because it is also the energy "cost" of making a unit area of a surface (or interface). Representative values of surface tension are given in Table 7.2.

7.2.2 Fluids in Motion

There are five attributes of the flow of fluids:

1. Flow can be *laminar/streamline/steady* or *turbulent/unsteady*. In laminar flow, a particle in the flow moves in a smooth manner along well-defined streamlines. In contrast, the motion is very random locally in turbulent flow. The Reynolds number *Re* is a dimensionless figure of merit that crudely divides the regimes of laminar and turbulent flow. It is the ratio between inertial force ($\rho u^2/2$; ρu^2 is used here) and viscous force ($\eta u/d$) per unit volume on the fluid, where ρ is the fluid density, u is the average speed of flow, d is the tube diameter, and η is the fluid *coefficient of viscosity* or the *dynamic or absolute viscosity*, which is defined later. This gives

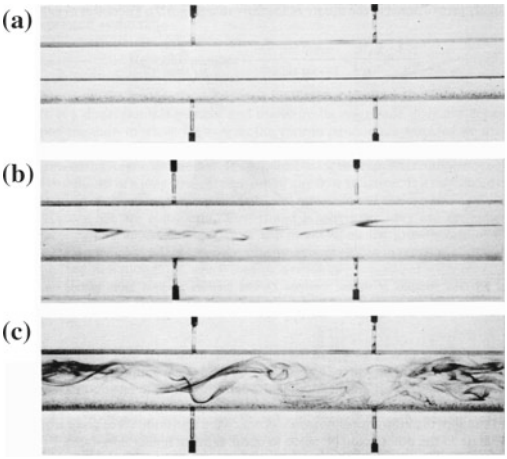


Fig. 7.5 Motion of a filament of dye in a straight pipe, showing (**a**) steady, laminar flow at low *Re*, (**b**) short bursts of turbulence for *Re* above the critical value, and (**c**) fully turbulent flow with random motion of the dye streak for higher *Re* (From [4]. Used with permission of Oxford University Press)

$$Re = \frac{\rho u^2}{\eta u/d} = \frac{\rho u d}{\eta} = \frac{ud}{\upsilon}, \tag{7.11}$$

where $\upsilon = \eta/\rho$ is the coefficient of *kinematic viscosity*.

Although this dividing line is not hard and fast, generally, flow in a rigid tube with $Re < 2{,}000$ is laminar and that with $Re > 2{,}000$ is turbulent. This dividing region is often cited as being between 1,200–2,500, and in the higher part of this range for smoother-walled tubes. Figure 7.5 shows flow in the laminar and turbulent regimes, and in the transition region between them.

2. Flow can be *compressible* or *incompressible*. Gases, such as air, are very compressible. Liquids are less compressible, and are often approximated as being incompressible.
3. Flow can be *viscous* or *nonviscous*. Fluids (other than superfluids) always have some viscosity, but in some cases it can be ignored totally, or first ignored and then considered as a perturbation.
4. Flow can be *rotational* or *irrotational*. In the cases we will consider there is no local rotation (such as vortices), so the flow will be irrotational.
5. Flow can be *steady* (constant in time) or *pulsatile* (with pulsing changes). Blood flow in the body is pulsatile, but is commonly treated as being in steady state in simple models. We will use both steady and pulsatile models in Chap. 8.

7.2.3 *Equation of Continuity*

The equation of continuity is a statement of the conservation of mass during flow. As seen in Fig. 7.6, when a fluid of a given mass density ρ moves with average speed u in a tube of cross-sectional area A, the product $\rho A u$ is constant (i.e., it is conserved).

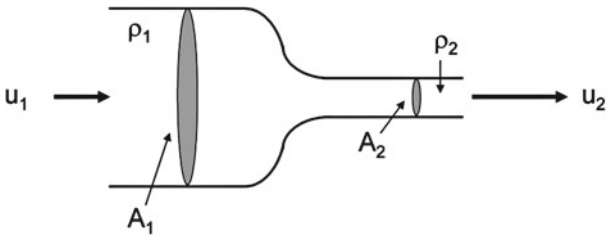


Fig. 7.6 Continuity of flow when the tube cross-sectional area changes

Because the speed is a longitudinal distance per unit time, Au is the volume flow per unit time (because $A \times$ distance = volume). Consequently, ρAu is the mass per unit time. In steady state, the same mass flows into a volume and leaves it. For the regions marked 1 and 2 in Fig. 7.6, this means that

$$\rho_1 A_1 u_1 = \rho_2 A_2 u_2. \tag{7.12}$$

If the fluid is incompressible, the densities in this equation do not depend on pressure and are the same everywhere. With $\rho_1 = \rho_2$, we follow the volume or volumetric flow rate Q, which is now a constant. This means $Q_1 = A_1 u_1$ and $Q_2 = A_2 u_2$, and so the continuity equation becomes $Q = Q_1 = Q_2$ with

$$Q = A_1 u_1 = A_2 u_2. \tag{7.13}$$

7.2.4 Bernoulli's Equation

Bernoulli's Principle (or equation) relates the average flow speed u, pressure P, and height y of an incompressible, nonviscous fluid in laminar, irrotational flow (Fig. 7.7). At any two points

$$P_1 + \frac{1}{2}\rho u_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho u_2^2 + \rho g y_2. \tag{7.14}$$

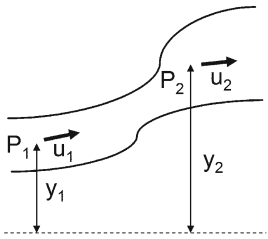


Fig. 7.7 For irrotational and nonviscous flow, the pressure, flow speed, and height are related by Bernoulli's equation along any streamline

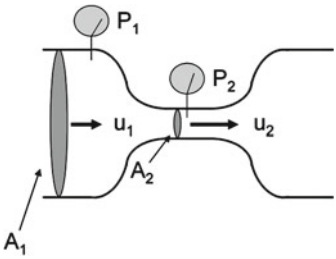


Fig. 7.8 Flow in a tube when the tube cross-sectional area changes. This is a Venturi tube, for which pressure and flow speed are related by Bernoulli's equation in the limit of constant height

The densities $\rho_1 = \rho_2 = \rho$ for this incompressible fluid. (Bernoulli's equation actually applies to any two points along a streamline.)

There are three special cases of Bernoulli flow. (1) For static fluids ($u = 0$), and Bernoulli equation's reduces to $P_1 + \rho g y_1 = P_2 + \rho g y_2$. (2) It reduces to Torricelli's theorem when $P_1 = P_2$, namely $\rho u_1^2/2 + \rho g y_1 = \rho u_2^2/2 + \rho g y_2$. (3) It reduces to the Venturi flow regime when $y_1 = y_2$ (Fig. 7.8), so

$$P_1 + \frac{1}{2}\rho u_1^2 = P_2 + \frac{1}{2}\rho u_2^2. \tag{7.15}$$

Because the continuity of flow in such a Venturi tube is $A_1 u_1 = A_2 u_2$

$$u_2 = \frac{A_1}{A_2} u_1. \tag{7.16}$$

Therefore we find

$$P_1 + \frac{1}{2}\rho u_1^2 = P_2 + \frac{1}{2}\rho \left(\frac{A_1}{A_2} u_1\right)^2 \tag{7.17}$$

and

$$P_2 - P_1 = \frac{1}{2}\rho u_1^2 \left(1 - \left(\frac{A_1}{A_2}\right)^2\right). \tag{7.18}$$

With $A_2 < A_1$, we see that $u_2 > u_1$ and $P_2 < P_1$. This shows that the flow becomes faster and the pressure becomes lower in clogged blood vessels.

7.2.5 Interactions Among the Flow Parameters

Pressure P, volume V, and flow rate Q are all interrelated in flow through vessels, be it blood flow in the circulatory system or air flow in breathing [22, 27, 28]. *Resistance* R_{flow} is the pressure difference or drop ΔP along a vessel, such as from

$P_1 = P(x = 0)$ to $P_2 = P(x = L)$ for a vessel of length L, needed to cause a given flow rate Q

$$R_{\text{flow}} = \frac{\Delta P}{Q}. \tag{7.19}$$

Compliance C_{flow} (or volume compliance) is the local change in volume ΔV caused by a local change in pressure ΔP in a vessel

$$C_{\text{flow}} = \frac{\Delta V}{\Delta P}. \tag{7.20}$$

Sometimes, the *inertance* L_{flow} is also defined. It is the change in pressure along a vessel caused by a flow rate that changes with time

$$L_{\text{flow}} = \frac{\Delta P}{\partial Q/\partial t}. \tag{7.21}$$

We will now consider the physics behind resistance, compliance, and inertance separately. In the next two chapters we will usually be able to analyze flow in vessels assuming that either resistance or compliance dominates, but for some vessels both resistance and compliance need to considered (and we will do this in the next chapter). Inertance is not important when the flow is constant in time. We will usually examine the overall flow properties of specific lengths of vessels, using models with these above "lumped" parameters. Occasionally we will analyze cases in which the parameters are distributed per unit length, such as flow resistance per unit length for volumetric flow along an artery in Chap. 8, using "distributed" or "transmission-line" models, as described in Appendix D. Such models are sometimes needed to examine time-dependent flow along a vessel, such as caused by the heart beating. Appendix D gives the differential form of these distributed equations when the effects of resistance, compliance, and inertance must all be considered and describes an analog between fluid flow and electrical circuits.

7.2.6 *Resistance in Flow: Viscous Flow and Poiseuille's Law*

Bernoulli's equation would predict that the pressure does not change during flow if the tube cross-section and height do not change. This is true for an ideal, nonviscous fluid. Viscosity is the friction during flow. It is always present and causes the pressure to drop during flow.

The coefficient of (dynamic or absolute) viscosity η is formally defined in (7.22), which gives the tangential or shear force F required to move a fluid layer of area A at a constant speed v, in the x direction, when that layer is a distance y from a stationary plate (Fig. 7.9) [8, 10]

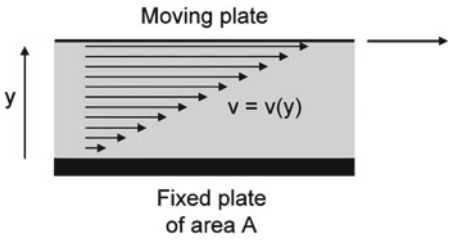


Fig. 7.9 Viscous fluid flow, with a linear gradient of fluid speed with position between a fixed and moving plate. This is shown for Newtonian flow

$$F = \eta \frac{A}{y} v. \tag{7.22}$$

This equation is also written as

$$\tau = \eta \frac{\mathrm{d}v}{\mathrm{d}y}, \tag{7.23}$$

where $\tau = F/A$ is the shear stress, as in (4.5) and Figs. 4.10 and 4.11, and $\mathrm{d}v/\mathrm{d}y$ is called the *shear rate*. (Check that the units of the shear rate are those that a rate should have, 1/s.) Fluids that are characterized by (7.22) and (7.23) are called "Newtonian fluids" and are said to undergo "Newtonian flow."

The SI units of η are $(\mathrm{N/m^2})\mathrm{s}$, which is equal to kg/m-s and Pa-s; this is called a Poiseuille (PI), but this unit is not often used. More commonly used is the poise (P) which is $10\times$ smaller. It is a natural unit in the CGS units system with $1\ \text{poise} = 1\,\mathrm{g/cm\text{-}s} = 0.1\ (\mathrm{N/m^2})\mathrm{s} = 0.1\,\mathrm{kg/m\text{-}s} = 0.1\,\mathrm{Pa\text{-}s}$. Also common is the centipoise (cP), with $1\,\mathrm{cP} = 0.01\ \text{poise} = 0.001\,\mathrm{Pa\text{-}s}$, because the viscosity of water at 20 °C is almost equal to 1 cP (and is actually 1.002 cP). We will usually use the units of Pa-s. Also, note that this viscosity coefficient is often called η by physicists (and is used as such here), whereas it is often called μ by biomedical engineers. It is also related to, but different from the viscosity damping constant for the dashpot c in (4.48).

Because of this drag, there must be a pressure difference (gradient) to maintain fluid flow in a tube. The relation between this pressure drop and the volumetric flow rate Q is given by Poiseuille's Law (or Hagen-Poiseuille's Law)

$$Q = \frac{\pi R^4}{8\eta L}(P_1 - P_2), \tag{7.24}$$

where R is the radius of the tube and L is its length (Fig. 7.10). This relation can be viewed as the flow rate for a given pressure drop. Alternatively, it can be viewed as the pressure drop when there is a flow Q in the tube

$$P_1 - P_2 = \Delta P = \frac{8\eta L}{\pi R^4}\, Q. \tag{7.25}$$

We will use this expression in Chap. 8 to determine the pressure drops in blood vessels during circulation. We will derive it soon as an advanced topic.

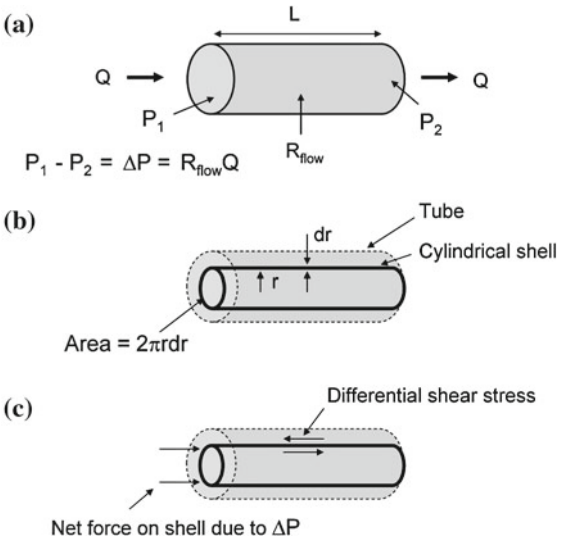


Fig. 7.10 Calculation of Poiseuille's Law for a tube in (**a**), using the cylindrical shell in (**b**), and balancing forces between the hydrostatic flow pressure force and the differential shear stress on the shell in (**c**)

Equation (7.25) is formally analogous to Ohm's Law for resistors, $V = IR_{\text{elect}}$ (or in a manner more parallel to this equation, $V = R_{\text{elect}}I$), where V is the voltage or potential difference across the resistor and is the driving term (which is analogous to ΔP), R_{elect} is the electrical resistance (analogous to the resistance of flow $8\eta L/\pi R^4$ here, which we will call R_{flow}), and I is the electrical current, which is the flow resulting from the driving term (analogous to the volumetric flow Q here).

Consider a tube with cross-sectional area A. The net force on the fluid in it is $(\Delta P)A$. If this force moves the fluid a distance L, the work done on it is $FL = (\Delta P)AL$. If this volume AL is moved in a given time, the work needed to do this in this given time—the power—is

$$P_{\text{power, flow}} = (\Delta P)Q, \tag{7.26}$$

or $P_{\text{power, flow}} = Q^2 R_{\text{flow}} = (\Delta P)^2/R_{\text{flow}}$. These expressions are analogous to those for the power dissipated by an electrical resistor: $P_{\text{power, elect}} = VI = I^2 R_{\text{elect}} = V^2/R_{\text{elect}}$.

The coefficient of viscosity for water is 1.78×10^{-3} Pa-s at 0 °C and it decreases with temperature, dropping to 1.00×10^{-3} Pa-s at 20 °C and 0.65×10^{-3} Pa-s at 40 °C. At 37 °C, η is 1.5×10^{-3} Pa-s for blood plasma and 4.0×10^{-3} Pa-s for whole blood, which are both higher than that for water at the same temperature. (Blood is really thicker than water.) The coefficients of viscosity of common human body fluids and other materials are listed in Table 7.3. As is clear from the table, the viscosity of liquids decreases with increasing temperature T, because the kinetic energy of molecules increases with T and this can overcome intermolecular forces that slow down motion between the dense, adjacent layers. In contrast, viscosity increases with temperature for gases, as $T(\text{in K})^{1/2}$, because diffusion between adjacent layers increases with T.

Table 7.3 Coefficient of viscosity η of common materials, in Pa-s (1 poise = 0.1 Pa-s)

Material	T (°C)	η
Water	0	1.78×10^{-3}
	20	1.00×10^{-3}
	37	0.69×10^{-3}
	50	0.55×10^{-3}
	100	0.28×10^{-3}
Blood plasma	37	1.5×10^{-3}
Whole blood[a]	37	$\sim 4.0 \times 10^{-3}$
Low shear rate, Hct = 45%		$\sim 100 \times 10^{-3}$
Low shear rate, Hct = 90%		$\sim 1{,}000 \times 10^{-3}$
High shear rate, Hct = 45%		$\sim 10 \times 10^{-3}$
Low shear rate, Hct = 90%		$\sim 100 \times 10^{-3}$
Cerebrospinal fluid	20	1.02×10^{-3}
Interstitial fluid	37	$1.0–1.1 \times 10^{-3}$
Human tears	37	$0.73–0.97 \times 10^{-3}$
Synovial fluid[b]	20	>0.3
Castor oil	20	1
Motor oil, SAE 10	20	0.065
Motor oil, SAE 50	20	0.54
Machine oil, heavy	37	0.13
Machine oil, light	37	0.035
Ethylene glycol	37	0.011
Mercury, liquid	37	1.465×10^{-3}
Methanol	37	0.47×10^{-3}
Ketchup	20	50
Peanut butter	20	250
Glass (anneal)	720–920 K	2.5×10^{12}
(blowing)	~1,300 K	$\sim 1 \times 10^{6}$
(furnace)	1,500–1,700 K	$\sim 1 \times 10^{2}$
Air	20	1.8×10^{-5}
	100	2.1×10^{-5}

Using data from [9, 15, 26]
Hct is the hematocrit, which is the volume fraction of red blood cells in blood
[a]See Figs. 8.10 and 8.11
[b]See Fig. 7.14

Derivation of Poiseuille's Law (Advanced Topic)

Now consider flow in a tube of radius R (Fig. 7.10). The distance radially from the center line of the tube is r. Using (7.22) and (7.23), the shear force and stress are

$$F = \eta A \frac{\mathrm{d}v}{\mathrm{d}r} \tag{7.27}$$

$$\tau = \eta \frac{\mathrm{d}v}{\mathrm{d}r}. \tag{7.28}$$

Imagine a series of concentric cylinders within this tube of thickness $\mathrm{d}r$ and length L (centered about the center symmetry axis, Fig. 7.10b), with a pressure drop ΔP along L. The force pushing one of these cylindrical shells forward is this pressure drop, ΔP, times the area of the front (and back) cylinder face, $2\pi r\mathrm{d}r$, or $(\Delta P)2\pi r\mathrm{d}r$. The viscous drag force that the cylindrical shell feels from the other shells (i.e., from the liquid) is the difference between the shear stress felt on its inner and outer surfaces $\times$ its surface area, $2\pi rL$. Using (7.28), this difference is

$$\frac{\mathrm{d}(2\pi rL\tau)}{\mathrm{d}r}\,\mathrm{d}r = \frac{\mathrm{d}\,(2\pi rL\eta\,\mathrm{d}v/\mathrm{d}r)}{\mathrm{d}r}\,\mathrm{d}r = 2\pi L\eta \frac{\mathrm{d}\,(r\,\mathrm{d}v/\mathrm{d}r)}{\mathrm{d}r}\,\mathrm{d}r \tag{7.29}$$

assuming the viscosity does not depend on r.

In steady state, the force due to the pressure drop plus the drag force equals zero, so

$$(\Delta P)2\pi r\mathrm{d}r + 2\pi L\eta \frac{\mathrm{d}\,(r\,\mathrm{d}v/\mathrm{d}r)}{\mathrm{d}r}\,\mathrm{d}r = 0 \tag{7.30}$$

or

$$\frac{\mathrm{d}\,(r\,\mathrm{d}v/\mathrm{d}r)}{\mathrm{d}r} = -\frac{\Delta P}{\eta L}\,r. \tag{7.31}$$

Integrating gives

$$r\frac{\mathrm{d}v}{\mathrm{d}r} = -\frac{\Delta P}{2\eta L}\,r^2 + C \tag{7.32}$$

$$\frac{\mathrm{d}v}{\mathrm{d}r} = -\frac{\Delta P}{2\eta L}\,r + \frac{C}{r}. \tag{7.33}$$

The constant C must equal zero, because otherwise the second term would be infinite at the center.

Integrating again gives

$$v(r) = -\frac{\Delta P}{4\eta L}\,r^2 + D. \tag{7.34}$$

(See Appendix C.) Because at the tube radius the velocity is zero ($v(R) = 0$), D is determined and this gives

$$v(r) = -\frac{\Delta P}{4\eta L}\,r^2 + \frac{\Delta P}{4\eta L}\,R^2, \tag{7.35}$$

so

$$v(r) = \frac{\Delta P}{4\eta L}\left(R^2 - r^2\right) = \frac{R^2 \Delta P}{4\eta L}\left(1 - \frac{r^2}{R^2}\right). \tag{7.36}$$

This speed is seen to be maximum in the center where $r = 0$. This maximum value of $R^2 \Delta P/(4\eta L)$ decreases to 0 as r increases from 0 to R.

The flow rate in the tube Q equals uA when the speed is uniform across the area A. When it is not, as here, Q is obtained by integrating $v(r)$ across the cross-sectional area. This is done by multiplying $v(r)$ by the area element $2\pi r\,\mathrm{d}r$ (the circumference $\times$ the differential in r) and integrating r from 0 to R, which gives

$$Q = \int_0^R \frac{\Delta P}{4\eta L}\left(R^2 - r^2\right) 2\pi r\,\mathrm{d}r \tag{7.37}$$

$$Q = \int_0^R \frac{\pi \Delta P}{2\eta L}\left(rR^2 - r^3\right)\mathrm{d}r = \frac{\pi \Delta P}{2\eta L}\left(\frac{R^4}{2} - \frac{R^4}{4}\right) = \frac{\pi R^4 \Delta P}{8\eta L}. \tag{7.38}$$

This is Poiseuille's Law $\Delta P = \left(8\eta L/\pi R^4\right) Q$ ((7.24) and (7.25)). Because Q is also equal to the area $\times$ the average speed, this average speed is

$$u = \frac{\pi R^4 \Delta P/8\eta L}{\pi R^2} = \frac{R^2 \Delta P}{8\eta L} \tag{7.39}$$

and

$$v(r) = 2u\left(1 - \frac{r^2}{R^2}\right). \tag{7.40}$$

This parabolic dependence is depicted in the rightmost profile shown in Fig. 7.11.

Many fluids are non-Newtonian fluids (Fig. 7.12), which means they are not characterized by (7.22) and (7.23), but by other relations [14, 25]. We assumed earlier that a fluid could generate no shear stress at any shear or strain rate; this is a frictionless or

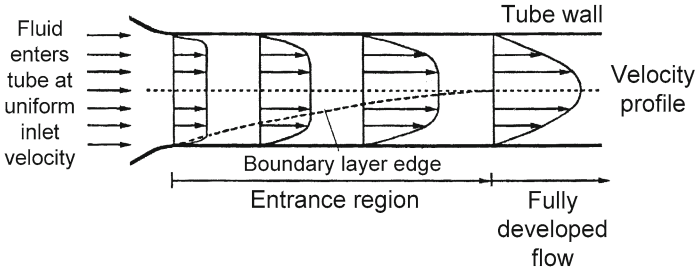


Fig. 7.11 Establishment of steady-state Newtonian flow into the parabolic velocity profile (in the fully developed flow) (From [9], based on [12]. Courtesy of Robert A. Freitas Jr., Nanomedicine, Vol. 1 (1999), http://www.nanomedicine.com)

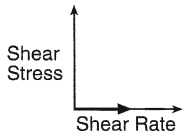
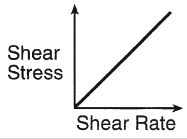
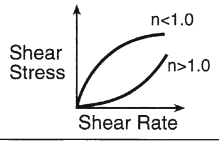
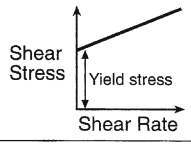
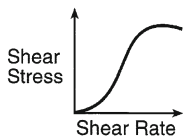
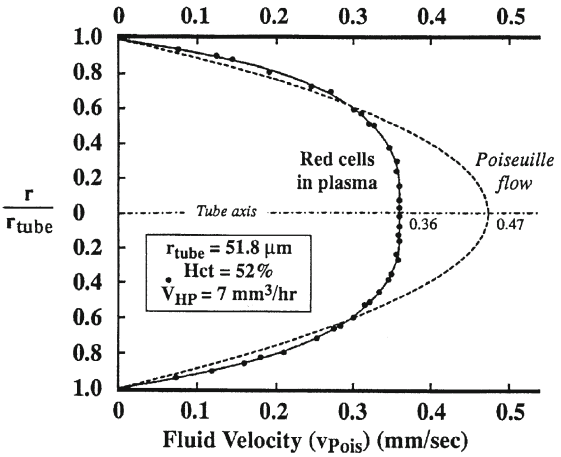
FLUID	SHEAR STRESS	PHYSICAL BEHAVIOR
Ideal	$\tau = 0$	
Newtonian	$\tau = \eta \frac{dv}{dy}$	
Non-Newtonian	$\tau = \eta \left(\frac{dv}{dy}\right)^n$	
Ideal Plastic	$\tau = \tau_y + \eta \left(\frac{dv}{dy}\right)$	
Viscoelastic	$\tau + \left(\frac{\mu}{\lambda}\right)\dot{\tau} = \eta \left(\frac{dv}{dy}\right)$	
τ_y = **yield stress** η = **coefficient of viscosity**		λ = **rigidity modulus** **n = constant**

Fig. 7.12 Newtonian and non-Newtonian fluid flow (From [14])

nonviscous fluid, which is unrealistic except for superfluids. (Note that such shear or strain "rates" are really gradients with respect to the direction normal to flow, i.e., y, and not with respect to time t. However, this terminology is reasonable because they have the same units as strain rates and because of the scaling argument given in Problem 7.23.) In some real non-Newtonian fluids, the shear stress is $F/A = \eta(\mathrm{d}v/\mathrm{d}y)^n$, where n could be greater or less than 1, as in Fig. 7.12. This is sometimes phrased as $F/A = \eta'(\mathrm{d}v/\mathrm{d}y)$ where the effective viscosity $\eta' = \eta(\mathrm{d}v/\mathrm{d}y)^{n-1}$ depends on the strain rate; as such a Newtonian fluid would have an effective viscosity that is independent of the strain rate. A dilatant or shear-thickening fluid has an effective viscosity that increases with increasing stress. A plastic or shear-thinning fluid has an effective viscosity that decreases with increasing stress. A Bingham plastic, such



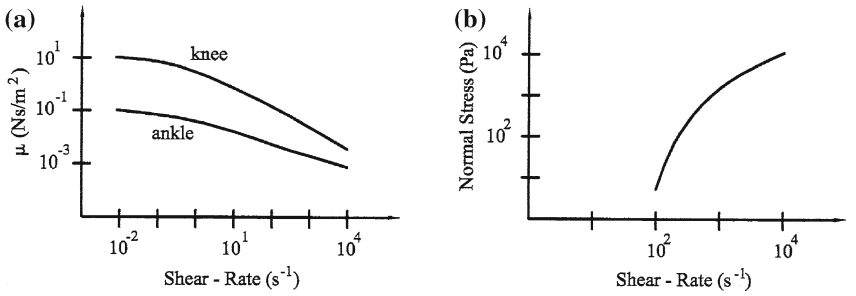


Fig. 7.13 Velocity flow profile of whole blood is blunted relative to the ideal parabolic flow of a Newtonian fluid (From [9], based on [12]. Courtesy of Robert A. Freitas Jr., Nanomedicine, Vol. 1 (1999), http://www.nanomedicine.com)

Fig. 7.14 **a** Synovial fluid is a non-Newtonian fluid, with a coefficient of friction that decreases with shear rate. **b** Another property of such a non-Newtonian fluid is that it can create a normal stress that depends on shear rate (From [21])

as toothpaste, has a finite yield stress even for $\mathrm{d}v/\mathrm{d}y = 0$, and above the yield stress it has a linear relationship with strain rate, $F/A = \alpha + \eta(\mathrm{d}v/\mathrm{d}y)$.

The composition of whole blood makes it a non-Newtonian fluid; this is discussed in Chap. 8, including in Fig. 8.11. Consequently, the flow pattern of blood is decidedly nonparabolic (Fig. 7.13). Synovial fluid is another example of a non-Newtonian fluid. Figure 7.14 shows that its coefficient of friction is high at low shear rates and much lower at high shear rates.

The dependence of flow on pressure drop within the laminar, intermediate, and turbulent regimes is shown in Fig. 7.15.

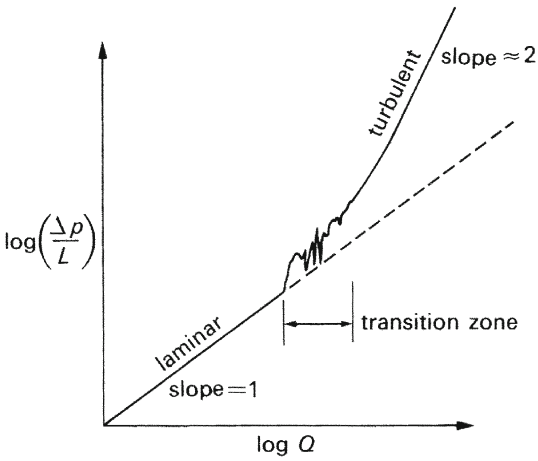


Fig. 7.15 Pressure drop per unit length versus log of the flow rate for a long tube, showing a transition region between laminar and turbulent flow (From [4]. Used with permission of Oxford University Press)

7.2.7 *Compliance in Flow*

We now determine the compliance C_{flow} by examining a thin-walled cylindrical tube of inner radius r, thickness w, and length L, with a pressure difference P between the inside and outside of the vessel (Fig. 7.16). The Law of Laplace for cylinders (7.4) shows that the tension T in the walls of a cylinder in equilibrium is rP. We can conceptually slit the vessel along its length and see that this tension (force per unit

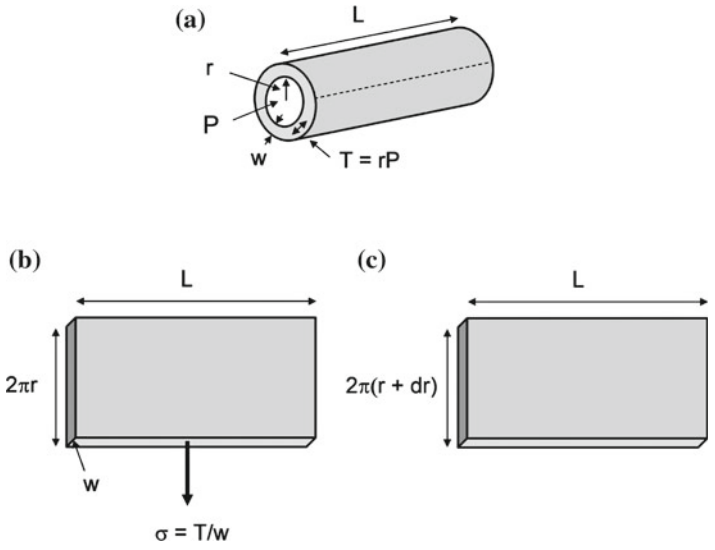


Fig. 7.16 Compliance vessels: unfolding the vessel. **b** Unfolded. **c** Effect of stress

length along L) corresponds to a force per unit area of T/w on the rectangular face with dimensions w and L. The length of the rectangular solid is $2\pi r$. Let us consider the stress–strain relation $\sigma = Y\epsilon$ in the context of this unfolded vessel. The stress is $\sigma = T/w$. The strain is $\epsilon = \mathrm{d}(2\pi r)/2\pi r = \mathrm{d}r/r = \Delta r/r$ or really $\Delta r/r_0$, where r_0 is the radius with no pressure. The stress–strain relation is

$$\frac{T}{w} = Y\frac{\Delta r}{r_0} \tag{7.41}$$

or with $T = rP$

$$\frac{\Delta r}{r_0} = \frac{T}{wY} = \frac{rP}{wY} = \frac{r_0}{w}\frac{P}{Y}. \tag{7.42}$$

The internal volume of the vessel is $V = \pi r^2 L$. Therefore we see that $\Delta V = 2\pi r(\Delta r)L$ and

$$\Delta V/V = 2\Delta r/r = 2(r_0/w)(P/Y). \tag{7.43}$$

For small changes in volume

$$V(P) = V_\mathrm{d}\left(1 + \frac{\Delta V}{V_\mathrm{d}}\right) = V_\mathrm{d}\left(1 + 2\frac{r_0}{w}\frac{P}{Y}\right) \tag{7.44}$$

$$= V_\mathrm{d} + 2V_\mathrm{d}\frac{r_0}{w}\frac{P}{Y} = V_\mathrm{d} + 2(\pi r_0^2 L)\frac{r_0}{wY}P, \tag{7.45}$$

with $V_\mathrm{d} = \pi r_0^2 L$. Using (8.5) the compliance $C_\mathrm{flow} = \Delta V/\Delta P = \Delta V/P$ is

$$C_\mathrm{flow} = \frac{2\pi r_0^3 L}{wY}. \tag{7.46}$$

In a more exact treatment, the right-hand side is multiplied by $1 - \upsilon^2$, as it is for the distensibility below in (7.51), where υ is Poisson's ratio [28]. This means that for $\upsilon = 0.5$, the 2 in this equation would be replaced by 3/2.

Distensibility

Such changes are equally well described in terms of the closely related distensibility D_flow of the tube. The cross-sectional area A of a tube increases by ΔA when the pressure difference between the inside and outside of the tube increases by ΔP. The distensibility is defined as the fractional change in area for a change in pressure:

$$D_\mathrm{flow} = \frac{\Delta A/A}{\Delta P}. \tag{7.47}$$

With $A = \pi r^2$ and $\Delta A = 2\pi r_0\,\Delta r$, we see that $\Delta A/A = 2\,\Delta r/r_0$ and using (7.42),

$$D_{\text{flow}} = \frac{2\,\Delta r/r_0}{\Delta P} = \frac{2(r_0/w)(\Delta P/Y)}{\Delta P} = \frac{2r_0}{wY} = \frac{1}{Y(w/d_0)}, \tag{7.48}$$

where $d_0 = 2r_0$ is the initial diameter, w is the wall thickness, and w/d_0 ($\ll 1$).

A more exact analysis relates the Young's modulus for circumferential stretch, which we still call Y, to the external and internal diameters d_e and d_i, the change in external diameter Δd_e occurring with this change in pressure difference, and Poisson's ratio υ [4]. This gives

$$Y = \frac{\Delta P}{\Delta d_\text{e}}\,\frac{2d_\text{e}d_\text{i}^2}{d_\text{e}^2 - d_\text{i}^2}\,(1-\upsilon^2). \tag{7.49}$$

For a thin-walled tube with wall thickness $w = (d_\text{e} - d_\text{i})/2 \ll d_\text{i}$, and with $d_\text{e} \sim d_0$ and $d_\text{i} \sim d_0 - 2w$, we find

$$Y = \frac{\Delta P}{\Delta d}\,\frac{d_0^2}{2w}\,(1-\upsilon^2). \tag{7.50}$$

With $\Delta A/A = 2\,\Delta d/d_0$ [22]

$$D_{\text{flow}} = \frac{\Delta A}{A}\,\frac{1}{\Delta P} = \frac{2\,\Delta d}{d_0}\,\frac{1}{\Delta P} = \frac{(1-\upsilon^2)}{Y(w/d_0)}. \tag{7.51}$$

This reduces to (7.48) for small Poisson's ratios.

7.2.8 *Inertance in Flow*

A decrease in pressure along a vessel can occur due to fluid viscosity; it can also occur if the fluid in it accelerates. Consider a plug of fluid of length L and area A, which has mass ρAL, where ρ is the mass density. The momentum of this plug is ρALu, where u is the flow speed. The momentum change with time, $\rho AL(\mathrm{d}u/\mathrm{d}t)$, is due to the force difference across the plug. In a vessel this equals $(P_1 - P_2)A$. The pressure drop across the distance L is then

$$P_1 - P_2 = \Delta P = \rho L(\mathrm{d}u/\mathrm{d}t) = (\rho L/A)\frac{\partial Q}{\partial t} = L_{\text{flow}}\frac{\partial Q}{\partial t}, \tag{7.52}$$

with $Q = Au$ and the last two forms written more accurately with partial derivatives. So, the inertance is [28]

$$L_{\text{flow}} = \rho L/A = \rho L/(\pi r^2) \tag{7.53}$$

for a vessel with radius r. A more accurate expression for the inertance, $L_{\text{flow}} = 9\rho L/4A$, can be obtained by assuming a flow pattern that is closer to the idealized parabolic pattern than the uniform flow profile we just assumed [27, 28].

7.2.9 *Flow Under Special Conditions*

Approach to Steady Flow

The results of Sect. 7.2.6 apply to steady, laminar flow. If a tube bifurcates—such as in branching arteries, the velocity profile we derived with its *boundary layer* at the tube circumference (where the flow velocity decreases to zero), will not represent the flow distribution immediately after the bifurcation. It will be valid only after a distance past the bifurcation called the entrance length, X [4]. Experimentally

$$X = 0.03d(Re) \tag{7.54}$$

for a straight pipe, where d is the diameter ($d = 2R$). For the laminar flow regime with $Re < 10$, this is not valid and the entrance length is shorter; when $Re \ll 1$ and inertial forces can be ignored

$$X \sim d. \tag{7.55}$$

For $Re > 2{,}500$, the flow is likely turbulent and the entrance length (for steady state turbulent flow) is shorter than that for fast laminar flow

$$X = 0.693d(Re)^{1/4}. \tag{7.56}$$

The development of parabolic flow for a Newtonian fluid in the laminar flow region is illustrated in Fig. 7.11.

Flow in Curving Tubes Such as Arteries

When you hold a hose with flowing water and try to change its direction you feel a resistance. This resistance is the force you need to apply to change the direction of the momentum of the water flow. This centripetal force becomes larger with faster flow rates (i.e., for larger hose areas and faster water flow speeds), as is well known to all firepersons. Curving arterial walls, such as the aorta, feel a pressure due to the difference in hydrostatic pressure inside and outside the vessel that arises from this force. This pressure is felt equally around the wall.

Consider a tube or artery of inner radius R that is turning with a radius $\mathcal{R}$. Figure 7.17 shows an arc of angle θ (in radians) $\ll 1$ of such a vessel. The average speed of flow is u and the mass density of blood is ρ. The magnitude of the momentum in the vessel per unit volume is $\rho u(AL)$, for a vessel with cross-sectional area $A = \pi R^2$ for a length L of blood flow. In traversing an angle θ, the momentum vector changes by $\sim(\rho u)(AL)\theta$. This occurs when the blood moves a distance

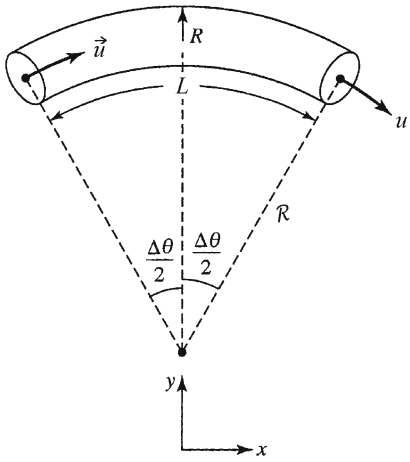


Fig. 7.17 Flow in a curved tube (From [2])

$\mathcal{R}\theta$, given by the arc length, with a speed u, so this occurs in a time $\mathcal{R}\theta/u$. Consequently, the force needed to do this is the change of momentum per unit time, which is $(\rho u AL\theta)/(\mathcal{R}\theta/u) = \rho u^2 AL/\mathcal{R}$. Because the mass of this volume of blood is $m = \rho AL$, this looks like the centripetal force $mu^2/\mathcal{R}$. (It looks like it, because that is what it is.)

This force is distributed across the outer half of the inner arterial wall, which has a cross-sectional area πRL. Because the force is outward, there is a larger load on the outermost portions shown in Fig. 7.17 and a smaller load on the outer upper and lower regions. Therefore, the peak force per unit area is more accurate when you use a smaller effective area, say $\pi RL/2$. Consequently, the peak pressure is the force per unit area $\rho u^2 AL/\mathcal{R}$, with $A = \pi R^2$, divided by this area $\pi RL/2$, or

$$P_{\text{cent}} = \frac{\rho u^2(\pi R^2)L/\mathcal{R}}{\pi RL/2} = 2\rho u^2 \frac{R}{\mathcal{R}}. \tag{7.57}$$

7.2.10 *Flow of Objects in Fluids: Drag and Lift*

The viscosity of a fluid also creates a drag force on objects that move in the fluid [1]. The reason for this is clear from (7.22); such objects are just like the plate in Fig. 7.9 in this functional definition of viscosity. Viscosity causes the boundary layer of the fluid near the ball (or plate) to move with it. If the object is moving at a speed u relative to the fluid, this drag force on the object is given by *Stokes Law*

$$F_{\text{drag,Stokes}} = 6\pi R\eta u, \tag{7.58}$$

where R is the hydrodynamic radius of the object, which is about half the typical lateral dimension D. This expression for Stokes friction is valid when the flow speed is slow enough that the streamlines about it are laminar. Here this means the Reynolds number $Re = \rho Du/\eta = 2\rho Ru/\eta$ is smaller than $\sim$100.

For Reynolds numbers much larger than 100, viscosity is no longer totally dominant and the main drag force is due to the formation of vortices that appear and trail the object, particularly as turbulent flow becomes important. This *hydrodynamic drag force* is

$$F_{\text{drag,hydrodyamnic}} = \frac{1}{2} C_{\text{D}} A \rho u^2, \tag{7.59}$$

where A is the frontal surface area and C_{D} is the drag coefficient. For $100 < Re < 2 \times 10^5$, $C_{\text{D}} \simeq 1.0$ for circular cylinders. For spheres, C_{D} decreases from 1.0 to $\simeq$0.5 as Re increases from 100 to 1,000 and it remains about 0.5 for $1{,}000 < Re < 2 \times 10^5$. For both cylinders and spheres, C_{D} becomes smaller at somewhat higher Re. These vortices or eddies are produced at the *Strouhal frequency*

$$f_{St} = \frac{(St)u}{D}, \tag{7.60}$$

where St is the Strouhal number. St depends on C_{D} and Re, and is typically between 0.12 and 0.23.

Problem 7.40 examines which drag regime dominates for human motion in fluids: walking and running in air and swimming in water.

Another source of drag that is present at all speeds is *skin friction*, which is due to the acceleration of the initially still fluid to the object speed u, because fluid in the boundary layer near the object sticks to it. This is different from Stokes drag, which is due to frictional losses in the fluid. This skin friction is

$$F_{\text{drag,skinfriction}} = \frac{1}{2} C_{\text{sf}} S \rho u^2, \tag{7.61}$$

where C_{sf} is the skin friction coefficient, which depends on the details of the flow, and S is the wetted surface area. When you swim at or near the surface, fluid builds up to a higher than ambient level in front of your head (as you push the water forward). The water is depressed to a level lower than ambient after your head, as it "ventilates." This *ventilation drag* force varies as u^4.

The power consumed by each of these drag forces is

$$P = F_{\text{drag}} u. \tag{7.62}$$

When a foil that is tilted up at an angle β moves in a fluid, an upward force is generated on it called *lift*, which is

$$F_{\text{lift}} = \frac{1}{2} C_{\text{lift}} S \rho u^2. \tag{7.63}$$

The lift coefficient, C_{lift}, varies linearly with this angle of attack. For small angles, it varies linearly from -0.4 to 1.2 for β varying from $-4°$ to $12°$ (for $Re = 1.7 \times 10^6$). Of course, $C_{\text{lift}} = 0$ for $\beta = 0°$. For β much larger than $12°$, the flow separates from the upper edge of the wing and there is stalling of the lift.

Chapter 3 discussed the lift force on spinning objects, such as thrown baseballs and such, which is commonly called the Magnus force.

7.3 Diffusion (Advanced Topic)

When the concentration of particles (or molecules) is not uniform, the random particle thermal motion leads to a net movement (or diffusion) of particles from regions of higher concentration to regions of lower concentration. The net effect is to make the concentration more uniform (Fig. 7.18). This diffusion flow rate increases with the nonuniformity or gradient of the concentration, which is *Fick's First Law of Diffusion*

$$J = -D_{\text{diff}} \frac{\partial n}{\partial x} \tag{7.64}$$

for flow in one-dimension, where J is the flux of particles (particle flow per unit area per unit time), D_{diff} is the diffusion coefficient, and $n(x, t)$ is the concentration of particles. (We must use partial derivatives here because everything depends on x and t.) The diffusion coefficient depends on the background medium, and is on the order of $\sim 10^{-1}\,\text{cm}^2/\text{s}$ in gas, $\sim 10^{-5}\,\text{cm}^2/\text{s}$ in liquid, and $\sim 10^{-9}\,\text{cm}^2/\text{s}$ in solid backgrounds.

During this flow the total number of particles must be conserved. Consider the cylindrical volume construct in Fig. 7.19, with its axis along the x-axis, and of length $\text{d}x$ and cross-sectional area A. The total number of particles entering from the left in a unit time $\text{d}t$ is $J(x)A(\text{d}t)$ and the number leaving from the right in this same time is $J(x + \text{d}x)A(\text{d}t) \simeq (J(x) + (\partial J/\partial x)\text{d}x)\, A(\text{d}t)$. Therefore the net increase in the number of particles in the cylinder is the difference $-(\partial J/\partial x)(\text{d}x)A(\text{d}t)$. This must be accounted for by the change in density in this time in the volume, which is $[(\partial n/\partial t)\text{d}t][A(\text{d}x)]$. This gives the conservation of the number of particles

$$\frac{\partial n}{\partial t} = -\frac{\partial J}{\partial x}. \tag{7.65}$$

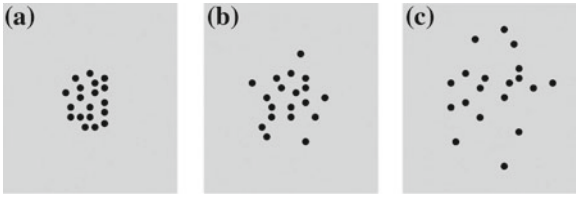


Fig. 7.18 Schematic of how the locations of particles vary at successively later times, from (**a**) to (**c**), as a result of diffusion

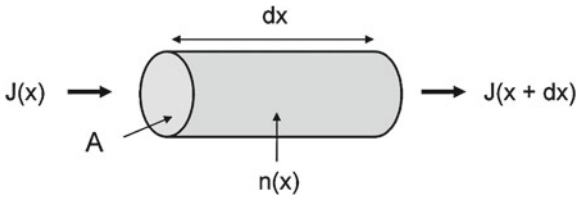


Fig. 7.19 Particles are conserved for any flow in and out of the cylinder through a change in concentration inside of it, for any flow process and for diffusion

These two equations can be combined by differentiating (7.64) in space to get $\partial J/\partial x = -D_{\text{diff}}\,\partial^2 n/\partial x^2$ (assuming that D_{diff} does not depend on x) and replacing $\partial J/\partial x$ from (7.65). This gives the *Diffusion equation* (or *Fick's Second Law of Diffusion*)

$$D_{\text{diff}}\frac{\partial^2 n}{\partial x^2} = \frac{\partial n}{\partial t}. \tag{7.66}$$

Such diffusion leads to a slow gaussian-like, undirected spreading of the species over a distance $x \sim \sqrt{2D_{\text{diff}}t}$ in a time t. (A gaussian profile is of the general form $\exp\left(-x^2/a^2\right)$.) For a total number of particles N initially at $x = 0$ at $t = 0$, the concentration is approximately

$$n(x,t) \sim \frac{N}{\sqrt{4\pi D_{\text{diff}}t}}\exp\left(-x^2/4D_{\text{diff}}t\right). \tag{7.67}$$

As presented, this solution is not valid for small times. The exact solution is slightly more complicated in other ways as well, but it is essentially the same result when the initial spread of particles is very small (Fig. 7.20). (See Appendix C.) If the initial distribution is gaussian, (7.67) becomes

$$n(x,t) = \frac{N}{\sqrt{2\pi\sigma^2(t)}}\exp\left(-x^2/2\sigma^2(t)\right), \tag{7.68}$$

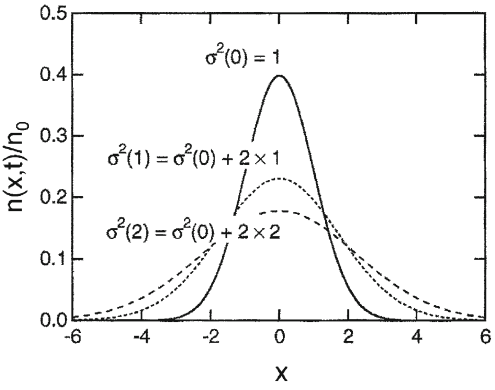


Fig. 7.20 Diffusion in one dimension, with gaussian spreading, with the initial distribution ($\sigma(0)$), one at time t during which σ^2 has tripled (in this particular example) ($\sigma(t)$), and one at time $2t$ ($\sigma(2t)$) (From [18, 19])

where

$$\sigma^2(t) = \sigma^2(0) + 2D_{\text{diff}}t \tag{7.69}$$

and $\sigma(0)$ is the initial spread.

In three dimensions, the spreading of particles by diffusion is described by

$$n(r,t) = \frac{N}{\left(2\pi\sigma^2(t)\right)^{3/2}} \exp\left(-r^2/2\sigma^2(t)\right), \tag{7.70}$$

with

$$\sigma^2(t) = \sigma^2(0) + 2D_{\text{diff}}t, \tag{7.71}$$

where $r^2 = x^2 + y^2 + z^2$.

Diffusion can be very important in the body over very small distances, on the order of 1–100 μm (~100 μm for oxygen diffusion), but is not very useful over much longer distances. The amount of material that can be transported from one place to another is limited by the lack of directionality of diffusion (Problem 7.26). It leads to an increase in disorder, whereas a functioning organism requires careful control and regulation within characteristic time frames. (The level or disorder is known as *entropy*, which is discussed in more detail in discussions of thermodynamics, statistical mechanics, and in several areas of biophysics.) The importance of diffusion in flowing systems is illustrated in Fig. 7.21 for an artery. Smelling objects involves the diffusion of molecules to your nose (Problem 7.30).

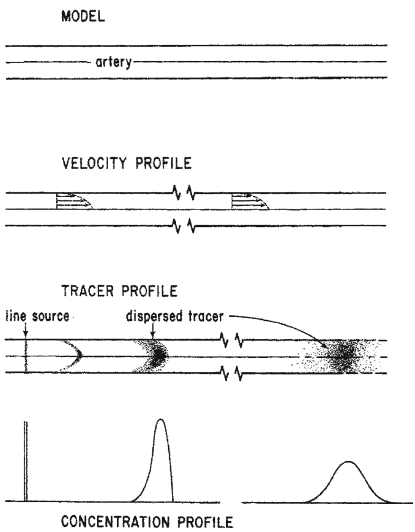


Fig. 7.21 Diffusion of an injected impulse, such as a dye, in an artery, with the shown line source initial distribution. The profile of the injection distorts as it adopts the velocity profile of the flow and it also diffuses (From [11])

7.4 Pressure and Flow in the Body

Table 7.1 gives characteristic pressures in the body. The blood pressure ranges from ~1–140 mmHg in different vessels and the speed of blood flow in these vessels ranges from ~0.05–50 cm/s. The overall volumetric flow rate is ~5 L/min. The relationship between pressure and flow in the circulatory system is detailed in Chap. 8. The characteristic pressure difference between the lungs and surrounding media is several mmHg and the volumetric flow rate of air into the lungs is ~6 L/min; this is discussed further in Chap. 9.

The flow rates in much of the human alimentary (digestive) system are quite slow (Table 7.4). Propulsive movements in this system are due to peristaltic action, with muscular contraction of the contractile ring around the gut sliding food forward, as diagrammed in Fig. 7.22. When there is a large amount of food in it, the gut stretches or distends and through sensors and feedback this stimulates a contractile ring 2–3 cm upstream. Mixing movements in the gut are caused by these peristaltic actions and by local constrictive contractions that occur every few cm in the gut and last for several seconds.

The relationship between volume and pressure is important in the digestive system. Pressure (tension) in the walls of the stomach increases during eating. The volume of the stomach of radius R increases as R^3. (This models the stomach as a sphere of volume V and ignores its finite radius with no food contents.) From (7.9), $\sigma = R(\Delta P)/2w = ((\Delta P)/2w)(3V/4\pi)^{1/3}$, so the tension in the stomach walls should increase, much slower, as R. Pressure in the stomach can also increase because of air swallowed during eating, which can lead to burping or belching. Bacterial action produces gas in the gut; at high enough pressure this causes flatulence.

As with the stomach, the pressure within the bladder increases slower than its volume, and this is seen in Fig. 7.23. The pressure rises to 5–10 cmH_2O when it is

Table 7.4 Approximate flow rates and other properties of the human alimentary system, estimated for a 70 kg male

Component	Length (cm)	External dimension or width (cm)	Internal volume (cm^3)	Luminal area (cm^2)	Contents passage time	Contents speed (cm/s)
Mouth and pharynx	8	2–5	~50	~80	1–10 s	1–8
Esophagus	25	1.3–2.5	~100	~200	5–20 s	3–5
Stomach	12	8	230–1,000	~600	2–6 h	~0.001
Small intestine	400	3–6	1,100	~3,500	3–5 h	0.03
Large intestine	~150	5.0–7.5	300	~2,000	10–20 h	0.004–0.008
Rectum	16–20	2.5–3.8	40	~100	~1 h	0.006
Total, average, or range	~600	~3.5	1,800–2,600	~6,500	16–32 h	~0.01

Using data from [9]

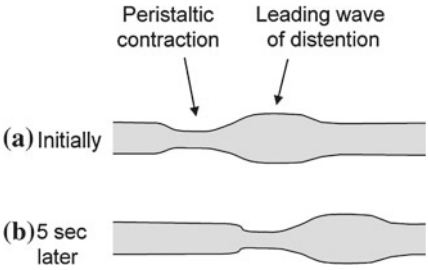


Fig. 7.22 Peristaltic action in the gut. (See Fig. 8.14 for peristaltic assistance in the return of venous blood to the heart) (Based on [13])

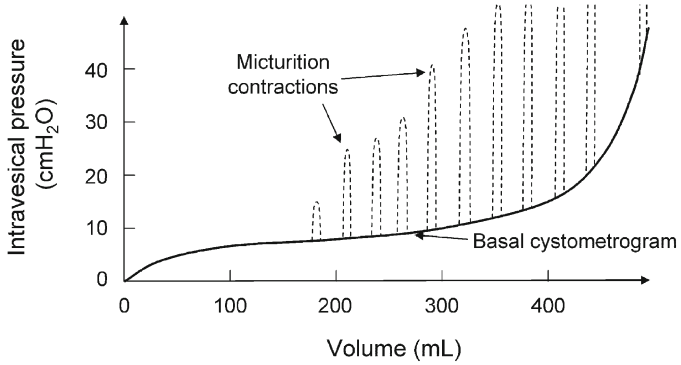


Fig. 7.23 Normal pressure–volume in the human urinary bladder (cystometrogram), also showing acute pressure waves (*dashed spikes*) caused by micturition reflexes (Based on [13])

filled by 30–50 mL of urine. (The units of cmH_2O are commonly used in this area, with $1\,cmH_2O = 0.738\,mmHg$.) Much additional urine can collect, 200–300 mL, with only a small rise in pressure. Above 300–400 mL the pressure increases rapidly. At $\sim$30 cmH_2O (3 kPa), there is an urge to urinate. Muscle contraction in the bladder (micturition reflexes) momentarily increases this pressure to 150 cmH_2O (15 kPa), with the normal voiding pressure being 20–40 cmH_2O (2–4 kPa). The wall tension increases with the volume of the bladder $V_{\text{bladder}}^{1/3}$, as seen from the Law of Laplace assuming a constant wall thickness. Therefore, the sensors to signal the urge to urinate would seem to be in the wall, sensing wall stress, and not sensing the pressure inside the bladder, because the pressure there is fairly constant.

7.5 Motion of Humans in Fluids

We have already encountered several examples of humans in fluids. One is the loss of heat by thermal conduction and convection to the surrounding air in Chap. 6. Drag is also important in walking, running, cycling, and so forth, as is clear from how wind

increases the metabolic needs during walking and running (Table 6.25). Locomotion in water, i.e., swimming, and potential human flight are examples in which the effects of the fluid are paramount [11].

7.5.1 Swimming

We are fairly buoyant, but not all can float. To float we must have an average density less than that of water ($1.0\,\mathrm{g/cm^3}$). (Equivalently, we must have a specific gravity (=density/water density) <1.) Those with relatively more fat (with endomorph body shapes) can float, with face, chest, and toes above the surface, because fat ($\simeq 0.8\,\mathrm{g/cm^3}$) has a density lower than water. Those who are relatively muscular or big-boned (mesomorphs) cannot float because the densities of muscle ($\simeq 1.0\,\mathrm{g/cm^3}$) and bone $\simeq 1.5$–$2.0\,\mathrm{g/cm^3}$ are, respectively, roughly equal to and larger that of water. People with an average density a bit higher than that of water may be able to float after taking in a deep breath because of the low density of air ($0.0012\,\mathrm{g/cm^3}$). Most men and women will float after taking in a deep breath, but most men will sink with just residual air in their lungs (after an normal exhalation, see Chap. 9). Very young and very old people are more likely to float because they have more fat, less muscle mass, and (for old people) lower long bone density. (Measuring body density and fat percentage is described in Problem 1.43.)

The center of buoyancy is the effective center of the buoyant forces and is determined by the differences in local body density and the density of water for the submerged part of the body. Since our legs are denser than our chest because of the low density of the lungs, when we try to back float with our bodies straight and horizontal, the centers of mass and buoyancy may not coincide or even lie on the same vertical line; the center of buoyancy can be closer to the head, especially for people with a low percentage of fat, and nearer the front of the body. This induces a torque on the straight body, and the body rotates until the center of mass is below the center of buoyancy (see Problem 7.31). Alternatively, we can adjust to cancel this torque by dropping our torsos into the water and dropping our legs even more so [16].

When floating (or almost floating) people push water parallel to the surface, in the "backwards" direction, they are propelled forward by the reaction force (Newton's Third Law). In other words, they swim. Because the arm and leg strokes are periodic, the forward propulsion is really periodic in theory, much like the periodic nature of blood rhythmically pumped by the heart. The net forward acceleration of the swimmer is due to the sum of this forward reaction response of the backward pushing of water and drag. (We are ignoring other lift forces [1, 5, 17].) For a person swimming with speed u

$$m_\mathrm{b}\frac{\mathrm{d}u}{\mathrm{d}t} = F_{\text{forward propulsion}} - F_\mathrm{drag}. \tag{7.72}$$

(More rigorously, $-F_\mathrm{drag}$ is really $-\mid F_\mathrm{drag} \mid$.) As is proved in Problem 7.40, the main source of drag is hydrodynamic, which scales as u^2, and not viscous

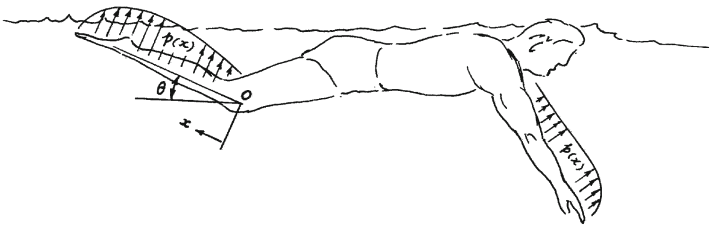


Fig. 7.24 Arm and leg motion during freestyle swimming (the crawl). Also see Problem 7.37 (From [11])

Stokes-type drag, which scales as u; consequently, $F_{\text{drag,hydrodynamic}} = \frac{1}{2} C_D A \varrho_{\text{water}} u^2$ (7.59). Actually, there are three identifiable sources of drag that scale as u^2: those due to frontal resistance, eddy resistance (due to water not filling in the body's wake and forcing the body to drag along these eddies), and surface drag. The first two types are sometimes collectively called hydrodynamic drag. Assuming now that this propulsion is continuous (as opposed to cyclic), the left side of (7.72) is zero and the steady state speed is

$$u^2 = \frac{2F_{\text{forward propulsion}}}{C_D A \varrho_{\text{water}}}. \tag{7.73}$$

For freestyle swimming (which is technically called the "crawl," Fig. 7.24), the propulsion force during a stroke can be estimated as the momentum gained by the pushed water during the duration of the stroke T_{stroke}, divided by that stroke time. (Remember, $F = ma$ can be expressed in terms of the momentum $p = mv$, as $F = \mathrm{d}p/\mathrm{d}t$ or $\Delta p/\Delta t$.) The momentum of the water is the mass of water displaced, $\varrho_{\text{water}} V_{\text{water}}$, times the final water speed, v_{water} (relative to the swimmer), or $\varrho_{\text{water}} V_{\text{water}} v_{\text{water}}$. Therefore we find

$$F_{\text{forward propulsion, stroke}} \simeq \frac{\varrho_{\text{water}} V_{\text{water}} v_{\text{water}}}{T_{\text{stroke}}}. \tag{7.74}$$

Let us consider forward propulsion due to motion of the hands and arms only and ignore leg motion. Let us also assume that the swimmer's arm is straight during the stroke and rotates about the shoulder in a cylindrical sweeping motion with a radius of the arm length l_{arm} and a width equal to the hand width w_{hand} (which we will say is also roughly equal to the arm width, w_{arm}). Therefore, the volume of displaced water is $V_{\text{water}} = \pi l_{\text{arm}}^2 w_{\text{hand}}/2$. (The factor of two accounts for the half of the cylindrical volume that is in the water.) The speed of the end of the arm is roughly $l_{\text{arm}}/T_{\text{stroke}}$ and near the shoulder it approaches zero. Because the water is moved at the arm speed, the average speed of the water is $v_{\text{water}} \simeq l_{\text{arm}}/2T_{\text{stroke}}$. (Part of the water is pushed downward during an ideal circular motion of the arm and this does not contribute to this forward propulsion; we will ignore this because the stroke motion is not really circular.) Therefore, (7.74) becomes

$$F_{\text{forward propulsion,stroke}} \simeq \frac{\varrho_{\text{water}} \pi l_{\text{arm}}^3 w_{\text{hand}}}{4T_{\text{stroke}}^2}. \tag{7.75}$$

We can estimate the arm length as the sum of the lengths of the upper and lower arms and half the length of the hand (because it is cupped), and so using Table 1.6 we see that $l_{\text{arm}} = 0.386H$, where H is the body height. We estimate that $w_{\text{hand}} = 0.07H$. Excellent swimmers make about 60 strokes a minute, so $T_{\text{stroke}} = 1\,\text{s}$. Using $H = 1.8$ m, we find that $F_{\text{forward propulsion,stroke}} \simeq 27\,\text{N}$. This is what we would expect for an effective force from a muscle with a cross-section of $1.3\,\text{cm}^2$ going into this motion, which seems a bit low. The steady state speed u is obtained from (7.73), using $C_\text{D} = 1.0$ and the transverse area $A \sim 0.076\,\text{m}^2$ (from the shoulder width, $0.259H$, times the chest depth, $0.09H$, using $H = 1.8\,\text{m}$). The average speed during this stroke is then $u \simeq 0.8\,\text{m/s}$.

The next stroke, with the other arm, starts when the previous one has stopped. The arm of this previous stroke "recovers" to the forward position above the water line and so it does not provide propulsion in reverse. Therefore, the forward propulsion is really continuous and this average speed seems reasonable. This speed of 0.8 m/s is not that different from typical swimming speeds and is not that far from the speeds of world-class freestyle swimmers. (The average speed for world-record men's freestyle swimming (in 2014) is ~2 m/s, decreasing from 2.39 m/s for 50 m distances to 1.82 m/s for 400 m.) Drag may be less than estimated here—in particular C_D and A may be smaller—and more water is likely being pushed per stroke by good swimmers than we estimated here. Remember that we totally ignored propulsion by the kick of the feet and legs and any propulsion by the rest of the body. Also, our analysis has ignored the complication of the initial dive into the pool and of reversing directions at the ends of the pool, etc.

The stroke is not exactly as described here. Actually, the arm does not pull straight in any stroke (freestyle (crawl), butterfly, breaststroke, and backstroke); after starting straight, it bends midway through and then (except for the breaststroke) straightens again for the crawl. This suggests that good swimmers use their hands more like propellers than paddles and that this type of motion can make lift significant, which we have ignored here.

The allometric relation for the swimming speed u of aquatic animals is

$$u \simeq 0.5 m_\text{b}^{0.19}, \tag{7.76}$$

where u is in m/s and m_b is in kg. This suggests that a 70 kg aquatic animal, such as a common dolphin, would swim at about 1 m/s, which is not far off from typical human performance. (Bottle nose dolphins have the same mass, but swim several times faster.)

What happens if a swimmer stops stroking and just glides? With no forward propulsion, (7.59) and (7.72) combine to give

$$m_\text{b} \frac{\text{d}u}{\text{d}t} = -\frac{1}{2} C_\text{D} A \varrho_{\text{water}} u^2. \tag{7.77}$$

Bringing the speed terms to the left gives

$$\frac{\mathrm{d}u}{u^2} = -\frac{C_\mathrm{D} A \varrho_\mathrm{water}}{2m_\mathrm{b}} \mathrm{d}t \tag{7.78}$$

and integrating from the initial speed u_i at $t = 0$ to the speed at time t gives

$$-\frac{1}{u(t)} + \frac{1}{u_\mathrm{i}} = -\frac{C_\mathrm{D} A \varrho_\mathrm{water}}{2m_\mathrm{b}} t. \tag{7.79}$$

Therefore the swimmer's speed approaches zero as

$$u(t) = \frac{u_\mathrm{i}}{1 + \frac{C_\mathrm{D} A \varrho_\mathrm{water} u_\mathrm{i}}{2m_\mathrm{b}} t} \tag{7.80}$$

with a characteristic time of say $18 m_\mathrm{b} / C_\mathrm{D} A \varrho_\mathrm{water} u_\mathrm{i}$, at which time $u = 0.1 u_\mathrm{i}$. (See Appendix C.)

Because $u = \mathrm{d}x/\mathrm{d}t$, we find

$$\mathrm{d}x = \frac{u_\mathrm{i}}{1 + \frac{C_\mathrm{D} A \varrho_\mathrm{water} u_\mathrm{i}}{2m_\mathrm{b}} t} \mathrm{d}t \tag{7.81}$$

Integrating from position $x = 0$ at $t = 0$ gives

$$x(t) = \frac{2m_\mathrm{b}}{C_\mathrm{D} A \varrho_\mathrm{water}} \ln\left(1 + \frac{C_\mathrm{D} A \varrho_\mathrm{water} u_\mathrm{i}}{2m_\mathrm{b}} t\right). \tag{7.82}$$

7.5.2 Human Flight

Why can we not fly? (That is, why cannot we fly without the assistance of a jet or helicopter, or propulsion devices on our backs?) The answer is easy. We cannot generate enough vertical force to counter our weight to enable us to hover or fly. In principle, we could do this by pushing air down fast enough or by generating a vertical force by aerodynamic lift—which could be possible if we could propel ourselves forward fast enough.

What happens if we try to fly by pushing air down by flapping our arms up and down? (Please do not try this in public.) The volume of air we could push down per arm flap is the arm area, which is length × width, times the distance pushed, which is approximately the arm length. This is roughly 0.7 m × 0.1 m × 0.7 m per arm or ∼0.1 m^3 for both arms. The mass density of air is 10^{-3} $\mathrm{g/cm}^3 = 1\,\mathrm{kg/m}^3$, so the mass displaced per flap is ∼0.1 kg. If the ends of our arms attained a speed of 80 mph ≃ 40 m/s (which is the speed of a fair major league fastball, and is clearly an overestimate), our average arm speed would be about 20 m/s. If we flapped our arms 3 times a second (which is also faster than expected), the change in momentum

in the moved air per unit time would be (0.1 kg)(20 m/s)(3/s) = 6 N (assuming no air is moved when our arms return to their initial positions at the end of each flap). This is much less than the weight of a 70 kg person, which is 700 N. Let us say we wear lightweight wings that would increase the effective flapping area to 2 m^2 (1 m^2 per wing) and the volume of the air we would move increases to 2 m^3. We would then generate an upward force of 120 N from this downward draft, and so even with our wildly high estimates of wing speed and flapping rate, we could not even approach developing enough vertical force to counter gravity and fly (or at least hover). (Because water has a density that is 1,000× that of air, we can easily keep ourselves afloat by pushing water down, and this is also assisted by buoyancy.) Such hovering, by the reaction force to the down draft in air, is more difficult than flying because there is no upward lift.

Perhaps we could flap and propel ourselves forward and develop some lift, or better yet: *Could we at least "takeoff" after running fast with our artificial wings in place?* Assume that a person accelerated to the world record speed of about 10 m/s and suddenly spread his or her 2 m^2 area wings. Using (7.63) under optimal conditions, we find $F_{\text{lift}} = \frac{1}{2} C_{\text{lift}} S \rho u^2 = (0.5)(1.2)(2\,\text{m}^2)(1\,\text{kg/m}^3)(100\ \text{m}^2/\text{s}^2)$, or 120 N of lift, which is still not enough. (Of course, even if the lift were enough and the person became airborne, forward deceleration due to drag would lead to a landing (or a crash).)

Clearly, any combination of wing flapping, for forward and some upward propulsion, and wing gliding for lift will also not lead to flight. The old saying, "If man (or woman) were meant to fly, he (or she) would have wings." is not true, because we could not fly even if we had wings. Of course people can hang glide with artificial wings; such gliding involves lift, drag, wind, and gravity.

Human-powered flight has indeed been demonstrated in the bicycle-powered aircraft built by the Paul MacCready team and cycled/flown by Bryan Allen, a champion bicyclist. In this aircraft the pedaling pilot propelled the propeller at the rear of the craft, which is connected to the cycle by a series of gears. Consequently thrust was created in this craft, which was optimized for lift, with minimal drag and weight. The "Gossamer Condor" flew for 7 min, 2.7 s in a closed course, and then on June 12, 1979 the "Gossamer Albatross" (with 30 m wingspan and 30 kg mass without the pilot) flew the first completely human-powered flight across the English Channel. It covered 35.6 km in 2 h 49 min, and thereby won the first awarded Kremer Prize, a prize established in 1959 to demonstrate human-powered flight. This world-class cyclist provided 125 W of mechanical power, flew very close to the surface to take advantage of the "ground effect"—which is a temperature inversion near the surface—and was completely exhausted at the end of the flight.

7.6 Summary

The Law of Laplace, the equation of continuity and Bernoulli's equation for nonviscous flow, and Poiseuille's Law of viscous flow can be used to model the flows of fluids in the body, such as blood and air—which are described in Chaps. 8 and 9, and

the movement of the body in fluids, such as swimming and flight, which is described in this chapter. The physics of pressure in fluids and diffusion are also used in these models.

Problems

Basic Fluidics and Pressure

7.1 Your blood pressure is measured with a sphygmomanometer, however with your upper arm pointed upward instead of downward. If your blood pressure is really 120 mmHg/80 mmHg, approximately what pressure would be measured?

7.2 The water level in a 4 m wide and 20 m long pool rises 0.75 mm when a person enters it and floats. What is the mass of that person?

7.3 You want to measure the volume of your whole arm by sticking it in an upright, long cylindrical tube with internal diameter of 15 cm, which is partially filled with water. The water level rises by 12.7 cm when a 50 kg female makes this measurement? What are the mass, weight, and volume of her arm? (See Chap. 1.)

7.4 Who is more buoyant and consequently floats higher: a large-boned, heavy muscled person with little body fat or a small-boned, lightly muscled person with more body fat?

7.5 Will retaining water affect a person's ability to float?

7.6 A 50 kg woman has a density of 1.01 g/cm^3 after normal exhalation. Does she float? Will she float after she inhales 2 L of air?

7.7 A 70 kg man with a density of 1.03 g/cm^3 ages. He gains 5 kg of fat. Will he float?

7.8 Will a person with an ectomorph shape float?

7.9 Why can all people float in the Dead Sea? (It has a specific gravity of 1.2–1.3. We have been assuming floating in water with no salt. Ocean water has a density of 1.027 g/cm^3.)

7.10 Three 50 kg women are airborne in a balloon filled with He. What is the minimum diameter of the balloon? (What assumptions are you making about the mass of the basket in which they are riding and the balloon itself?)

7.11 If the gauge pressure inside a cylindrical tube is 100 mmHg and its radius is 1 mm, what is the tension in the tube wall at equilibrium (in SI units)?

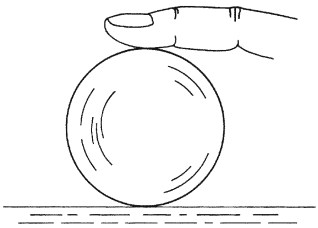


Fig. 7.25 Palpation of a blood vessel (From [10].) For Problem 7.14

7.12 (advanced problem) Derive the Law of Laplace for a sphere (7.9) by careful integration of the normal force on a hemisphere, in a manner analogous to the integration in (7.7) for a half-cylinder.

7.13 Over a large range of volumes, the pressure in the bladder is at a fairly constant value near 8 mmHg. If the thickness of the bladder is 5 mm, show that the wall tension is $\sigma = 600\,\text{Pa/cm}\ V_{\text{bladder}}^{1/3}$ where the bladder volume is in cm^3.

7.14 The internal pressure of an elastic vessel, such as an artery, vein, eyeball, aneurysm, or balloon, can be estimated by pushing down on it with your finger; this method is called *palpation* (Fig. 7.25):
(a) Show that the pressure felt by the finger is affected by the tension in the vessel wall.
(b) Show that the pressure you feel equals the pressure internal to the vessel when you push down on it so that the vessel wall is flat.

7.15 Assuming no viscosity and no changes in height, determine how the flow speed in a vessel changes if its diameter decreases by a factor of 4.

Viscous Flow

7.16 Compare the SI units of dynamic viscosity, η in (7.23), with those of the viscosity damping constant of the dashpot, c in (4.48).

7.17 One wants to use oil in car engines so the oil viscosity is a specific, optimized value—especially when the engine is started cold. Usually a heavyweight oil is used in very hot weather, such as SAE 50, and a lightweight oil in very cold weather, such as SAE 10. Using Table 7.3, estimate the viscosity needed at moderate temperature. Also estimate how the motor oil viscosity changes with temperature. (Nowadays, multiviscosity oils, such as SAE 10W/40 are used, which are suitable over a wide range of temperatures.)

7.18 Viscous flow with flow rate Q in a big tube of diameter D and length L, subdivides into N identical small tubes of length L with equal flow rates:
(a) What is the flow rate in each small tube?
(b) You are told that the pressure drops across the big tube and across the small tubes are the same (and both equal to ΔP). Find the diameter of the small tubes and determine if this is possible.

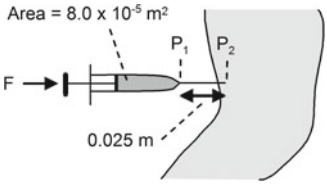


Fig. 7.26 Intravenous injection by a hypodermic syringe (Based on [6].) For Problem 7.20

(c) If instead, the diameters and lengths of the small tubes are $\alpha\times$ and $\beta\times$ that of the big tube, what is the resistance across each small tube and across the whole small tube system in terms of the resistance across the big tube?

7.19 We are very sensitive to even small changes in core body temperature. Let us examine what happens when the viscosity of blood changes because of such temperature changes. It is known that the dynamic viscosity of whole blood decreases by 30% when temperature increases from 25 to 37 °C. What is the increase in systolic blood pressure, from its normal value of 120 mmHg, needed to pump blood throughout the body at the same rate if the core body temperature decreased to 25 °C, with everything else being the same? (This temperature change will affect the body in many other ways even more dramatically; see Chap. 13.)

7.20 (a) How much force F (in N and lb) must be applied to a plunger to inject $1.0 \times 10^{-6}\,\text{m}^3$ of the solution in 3.0 s with a hypodermic syringe? Apply Poiseuille's Law for the pressure drop across the needle, as in Fig. 7.26 [6]. The needle is injected into a vein with a (gauge) pressure of 14 mmHg (1,900 Pa). Assume the plunger has an area of $8.0 \times 10^{-5}\,\text{m}^2$ and the syringe is filled with a solution with viscosity of 1.5×10^{-3} Pa-s. The needle has an internal radius of 4.0×10^{-4} m and a length of 0.025 m. Remember that you want to apply a (gauge) pressure in excess of the venous pressure to achieve the desired flow rate Q.
(b) Why are such injections performed intravenously and not intra-arterially?

7.21 Calculate the Reynolds number for each component of the alimentary system. Assume the coefficient of viscosity is 1 N-m/s to the stomach and 10 N-m/s after the stomach. Is the flow streamline or turbulent?

7.22 (a) For Newtonian flow, calculate the shear stress on the wall of a tube of radius R, for an average fluid speed u and fluid viscosity η.
(b) Estimate this (in SI units) for a typical human artery.

7.23 (advanced problem) Show that the strain rate $\text{d}v/\text{d}y$ used in flow is related to the time rate of change of strain $\text{d}\epsilon/\text{d}t$. (Hint: Express strain as the partial derivative of a deformation u, $\epsilon = \partial u/\partial y$ and speed as $v = \partial u/\partial t$. Then evaluate $\partial\epsilon/\partial t$, and switch the order of the y and t derivatives.)

Diffusion

7.24 Important molecules are formed in the middle of a 2 μm-diameter cell. How long does it take for them to diffuse throughout the cell? (Assume the cell contents are liquids and that the diffusion coefficient $D_{\text{diff}} = 10^{-5}\,\text{cm}^2/\text{s}$.) Is this fast enough to achieve normal metabolic activity rates?

7.25 In one-dimension, estimate the characteristic distances for diffusion in 1 s in a gas, liquid, and solid.

7.26 A $1\,\text{mm}^3$ volume of biological material must be transported 2 cm away to another $1\,\text{mm}^3$ region:
(a) If it flows in a vessel at a speed of 10 cm/s, how fast does it get there and what fraction of it arrives there?
(b) If it diffuses in a liquid with $D_{\text{diff}} = 10^{-5}\,\text{cm}^2/\text{s}$, approximately when will the maximum amount of it arrive and approximately what fraction of it will arrive?
(c) Which mode of transport is preferred and why?

7.27 Refer to Fig. 7.20. If σ and the abscissa are in cm and t is in s, what is D_{diff}?

7.28 Use substitution to confirm that (7.68) and (7.69) are the solution to the one-dimensional diffusion equation, (7.66).

7.29 (advanced problem) Use substitution to confirm that (7.70) and (7.71) are the solution to the three-dimensional diffusion equation

$$D_{\text{diff}} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial n}{\partial r} \right) = \frac{\partial n}{\partial t}. \tag{7.83}$$

7.30 (a) You can detect 4×10^8 molecules of ethyl mercaptan (which causes the rotten fish smell) per cm^3, which corresponds to approximately one molecule per 10^{11} molecules in air (because the air density is $5 \times 10^{19}/\text{cm}^3$). If $1\,\text{mm}^3$ of this liquid is released 10 m away, how long will it take to notice this release? (Ethyl mercaptan, C_2H_5SH has 62.1 g/mole and is a liquid with a density of $1.01\,\text{g/cm}^3$. It has an odor resembling that of rotten eggs, and is added to natural gas and propane to give those normally odorless fuels a distinctive smell.)
(b) If your dog's nose is a thousand times more sensitive, when will she or he smell it?

Swimming, Flying, and Drag Forces

7.31 Draw a diagram showing a person straight and floating on her back, with the locations of the centers of buoyancy and mass, first with a level body and then after rotation due to torques on the body.

7.32 Repeat the analysis that determines the speed of a swimmer, but now assume that the effective force of 2 in diameter muscles is providing 405 N continuously. (Why is this force reasonable?) Does your answer make sense? Why?

7.33 Repeat the analysis of the speed of a swimmer, but now assume that Stokes friction is the only dominant drag force. Does your answer make sense? Why?

7.34 Go through all the steps in determining the position during gliding in swimming, from (7.81) to (7.82).

7.35 Using the parameters in the text, estimate the characteristic time needed for a world class freestyle swimmer who stops stroking and glides to slow down. Also estimate the distance she travels in that time.

7.36 Repeat the analysis of gliding, (7.77)–(7.82), assuming only Stokes drag.

7.37 In Fig. 7.24, the lower leg of a swimmer is hinged at the knee (at $x = 0$) and is acted on by forces that are normal to its axis with force per unit length of $p(x)$. Show that the work done by the leg, of length L, as is rotates by $\mathrm{d}\theta$ is [11]

$$\mathrm{d}W = \left(\int_0^L p(x) x \mathrm{d}x \right) \mathrm{d}\theta. \tag{7.84}$$

7.38 Could people fly on another planet using artificial wings? How would g and the mass density of the atmosphere ρ have to change? (Does a change in g imply the same or an oppositely signed change in ρ?) Ignore the impact of spacesuits, differences in temperature, changes in metabolism, muscle atrophy, and so on.

7.39 What is the Reynolds number of a piece of matter 1 μm in diameter, such as a cell in water or particulate in blood? Assume a density of 1 g/cm^3, a speed of 4 mm/s, and the viscosity of blood. What type of drag dominates?

7.40 (a) Is Stokes friction or hydrodynamic drag dominant for people walking and running in air?
(b) Which is dominant for people swimming in water?
(Make sure you calculate the Reynolds number in each case.)

7.41 (a) Estimate the hydrodynamic drag force on a very fast runner.
(b) How much power is lost to drag?
(c) How does this compare to the metabolic power needed for running?

7.42 Speed skaters often adopt a position with a nearly horizontal trunk and downhill skiers adopt the "egg" position with a hunched-down body and skis pointed backward when they are not maneuvering. Why?

7.43 A person without a parachute is dropped from a plane at an altitude of 1,000 m. Determine the "terminal" speed of the person by equating the forces of gravity and drag. (Which drag limit is appropriate? Is there enough time for a constant final speed to be attained?)

7.44 (a) A person with a parachute is dropped from a plane at an altitude of 1,000 m. Determine the final steady state speed of the person by equating the forces of gravity and drag. (Which drag limit is appropriate? Is there enough time for a constant final speed to be attained?) Assume the person has a mass of 70 kg and the parachute has a negligible mass and is 7 m across when it is open.
(b) What is the minimum height above ground that the parachute should be opened so the person lands with a speed no greater than 1.5 m/s? Assume it takes 2.5 s for the parachute to deploy fully.

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